

Multiple Choice

1. **Answer:** A

Options: A) $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$; B) $\text{PbH}_4 < \text{SnH}_4 < \text{CH}_4 < \text{GeH}_4 < \text{SiH}_4$; C) $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$; D) $\text{CH}_4 < \text{PbH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{SiH}_4$

Note: Correct option: A - $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$

2. **Answer:** D

Options: A) 1.5 R; B) 2 R; C) 2.5 R; D) 3.5 R

Note: Correct option: D - 3.5 R

3. **Answer:** B

Options: A) 0.0471 Hz; B) 21.2 Hz; C) 42.4 Hz; D) 66.7 Hz

Note: Correct option: B - 21.2 Hz

4. **Answer:** D

Options: A) I only; B) II only; C) III only; D) I and III only

Note: Correct option: D - I and III only

5. **Answer:** D

Options: A) Na_2CO_3 ; B) Na_3PO_4 ; C) Na_2S ; D) NaCl

Note: Correct option: D - NaCl

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3.74 T'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '17 O'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 5.18 mT. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $\text{HBr(g)} \rightarrow \text{H(g)} + \text{Br(g)}$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key